

QUANG-HUY NGUYEN

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RESEARCH INTERESTS

Computational Biology: Multi-omics integration, pathway analysis, bulk and single-cell genomics, genome-wide association studies, polygenic risk prediction, and spatially resolved transcriptomics, metabolomics, and proteomics. **Computational Healthcare Informatics:** Data-driven methods for learning from heterogeneous clinical and biological data to support medical decision-making and public health. **Artificial Intelligence Foundations:** Multi-modal modeling, agentic reasoning systems, uncertainty quantification for reliable and interpretable AI, and federated and collaborative learning frameworks for privacy-preserving and distributed intelligence.

EDUCATION

Auburn University

Auburn, AL

Ph.D. in Computer Science & Software Engr - cGPA: 3.73/4.0

Aug 2023 – May 2027 (Expected)

Advisor: Professor Wei-shinn Ku.

- Creator of *scBatchProx*: Designed and implemented a federated latent-space framework for single-cell batch correction, delivering 3–8% embedding quality gains, with consistent improvements in batch correction and biological conservation across 85%+ of benchmarks. Led manuscript preparation and writing.
- Creator of *FedWQ-CP*: Designed and trained federated conformal CNNs for uncertainty quantification on Slurm-managed HPC clusters across 7 vision and medical imaging datasets, achieving robust client-level and global coverage under joint data and model heterogeneity. Led manuscript preparation and writing.

Dai Nam University

Hanoi, Vietnam

Bachelor of Pharmacy - cGPA: 8.09/10.00

Oct 2015 – Aug 2020

RESEARCH EXPERIENCE

Department of Computer Science, HUST

Hanoi, Vietnam

Visiting Graduate Student

Dec, 2024 – Jan, 2025

Advisor: Associate Professor Duc-Hau Le.

- Creator of *DeepSSC*: Designed and trained a semi-supervised multi-omics classifier on shared multi-GPU servers, evaluated across 3 TCGA cohorts, achieving 96.0% accuracy (99.2% AUC) and up to 19% gains over graph baselines. Co-mentored undergraduate researchers and contributed to manuscript preparation.

Department of Computer Science and Software Engineering, Auburn University

Auburn, AL

Graduate Research Assistant

Aug 2023 – Aug 2024

Advisor: Research Assistant Professor Phi Hung Bya and Associate Professor Tin Nguyen.

- Conducted a literature review on functional analysis methodologies in LC-MS-based targeted and untargeted metabolomics. Dockerized 13 surveyed software tools as services, simplifying installation and execution to enhance accessibility for readers with limited IT expertise.
- Conducted a literature review on large language models and their applications in biomedical domain.
- Led the *Candidozyma auris* project, performed BLAST-based gene annotation and pathway enrichment analysis to identify significant pathways involved in stress adaptation and antifungal resistance; contributed to manuscript proofreading.
- Led the type 2 diabetes (T2D) polygenic score project, developed and implemented a novel pipeline to derive gene-level signals from VCF genotypes for pathway analysis without matched gene expression data, identifying T2D-relevant pathways, and demonstrated pathway-specific polygenic risk scores achieved high AUC values; co-drafted the manuscript.

School of Computer Science and Engineering, Thuyloi University

Hanoi, Vietnam

Visiting Scientist

Oct, 2021 – Jul, 2023

Advisor: Associate Professor Duc-Hau Le.

- Creator of *DrGA*: Designed and implemented a modular toolkit in R tailored for the integrated analysis of cancer-associated genes. Led manuscript preparation and writing.

School of Computer Science and Engineering, Thuyloi University

Hanoi, Vietnam

Visiting Student

Dec, 2017 – Jul, 2019

Advisor: Associate Professor Duc-Hau Le.

- Read and replicated research papers, designed experiments and conducted algorithm on high-dimensional omics data in R.

INDUSTRY EXPERIENCE

Center for Biomedical Informatics, Vingroup Big Data Institute

Research Scientist

Hanoi, Vietnam

Oct 2020 – Sep 2021

Managers: Associate Professor Duc-Hau Le and Dr. Nam Sy Vo.

- Creator of *oCEM*: Designed and implemented a dual-purpose R package equipped with decomposition methods for (i) automatically detecting and analyzing overlapping co-expressed gene modules from a set of gene expressions and (ii) providing an auxiliary statistical approach to select the optimal number of principal components using a permutation matrix. Led manuscript preparation and writing.
- Designed and implemented a multi-omics analysis pipeline in R that detected novel prognostic subgroups of breast cancer, achieving a Cox log-rank p-value of 0.002 compared to the PAM50 classification scheme's p-value of 0.598. Led manuscript preparation and writing.
- Co-mentored an undergraduate student in clustering uveal melanoma patients in R.
- Contributed to the design and planning of the VN1K variant detection pipeline for large-scale genomic analysis of the Vietnamese population.

Research Intern

Mar 2020 – Oct 2020

Internship mentor: Associate Professor Duc-Hau Le.

- Developed and implemented a pipeline in R to systematically identify and analyze cancer driver genes using multi-omics data, ensuring a consistent approach to uncover key genetic factors contributing to cancer progression.

RESEARCH PROJECTS AND COMPETITIONS

Cell Maps for AI (CM4AI) CodeFest, UAB

Birmingham, AL

NIH Bridge2AI Consortium

Aug, 2025

- Collaborated in a team of 5 to create separate embeddings of yeast fitness data, proteomic mass spectrometry, and cellular imaging, then fuse them into a co-embedding to construct a with-yeast multiscale subcellular hierarchy that uncovered hidden biological processes in triple-negative breast cancer.
- Implemented the analysis pipeline based on the agreed workflow in Python.

The 1000 Vietnamese Genomes Project (VN1K)

Hanoi, Vietnam

Vingroup Joint Stock Company

2019 – 2024

PI: Associate Professor Duc-Hau Le and Dr. Nam Sy Vo.

GenomeAsia100K

Hanoi, Vietnam

NTU and others

2017 – 2022

PI: Dr. Stephan Schuster (Singapore side) and Associate Professor Duc-Hau Le (Vietnam side).

TEACHING EXPERIENCE

Department of Computer Science and Software Engineering, Auburn University

Auburn, AL

COMP-1220 & COMP-1230

Aug 2025 – Present

- Graduate Teaching Assistant for COMP-1220 An Intro to Programming in Python & COMP-1230 Introduction to Computing with Matlab.
- Teaching Python and Matlab fundamentals to ≈ 60 undergraduates and guiding them through problem-solving exercises in lab sections.

COMP-1230

Aug 2024 – May 2025

- Graduate Teaching Assistant for COMP-1230 Introduction to Computing with Matlab.
- Taught Matlab fundamentals to ≈ 60 undergraduates and guided them through problem-solving exercises in lab sections.

PUBLICATIONS

†co-first author, *co-corresponding author

Preprints & Preparations

4. Trang T. H. Tran, ..., **Quang-Huy Nguyen**, ..., Duc-Hau Le*, Quan Nguyen*, Van H. Vu*, Nam S. Vo*. *VN1K: a genome graph-based and function-driven multi-omics and phenomics resource for the Vietnamese population*. **bioRxiv**, 2025.04.15.648991. (In revision at **Nature Communications**). 2026
3. **Quang-Huy Nguyen**, Zongliang Yue, Hao Chen, Wei-Shinn Ku*, Jiaqi Wang*. *Federated-inspired single-cell batch integration in latent space*. **arXiv**, 2602.00423. (Under review). 2026

2. **Quang-Huy Nguyen**, Jiaqi Wang*, Wei-Shinn Ku*. *Conformalized Neural Networks for Federated Uncertainty Quantification under Dual Heterogeneity*. **arXiv**, 2602.23296. (Under review). 2026
1. Hoang Le†, Van-Minh Nguyen†, Duc-Hai Tran, Van-Anh Tran, Van-Nui Nguyen, **Quang-Huy Nguyen†***, Duc-Hau Le*. *A Semi-Supervised Deep Learning Framework for Multi-Omics Cancer Subtype Classification and Biomarker Discovery*. **bioRxiv**, 2022.01.13.476268. (Under review). 2026

Peer-reviewed publications

9. Ching-Lan Chang†, Michael A. Mosh†, **Quang-Huy Nguyen†**, ..., David Hess*, Edwin C. Oh*. *Wastewater intelligence predicts the emergence of clinically-relevant and drug-resistant *Candida auris* at healthcare facilities*. **medRxiv**, 2025.07.29.25332388. (Acceptance in Principle at Nature Communications). 2026
8. **Quang-Huy Nguyen**, Ha Nguyen, Edwin C. Oh, Tin Nguyen. *Current approaches and outstanding challenges of functional annotation of metabolites: a comprehensive review*. **Briefings in Bioinformatics**, 25(6):bbae498. 2024
7. **Quang-Huy Nguyen**, Tin Nguyen, Duc-Hau Le. *DrGA: cancer driver gene analysis in a simpler manner*. **BMC Bioinformatics**, 23:86. 2022
6. **Quang-Huy Nguyen**, Tin Nguyen, Duc-Hau Le. *Identification and validation of a novel three hub long noncoding RNAs with m6A modification signature in low-grade gliomas*. **Frontiers in Molecular Biosciences**, 9:801931. 2022
5. **Quang-Huy Nguyen**, Duc-Hau Le. *oCEM: Automatic detection and analysis of overlapping co-expressed gene modules*. **BMC Genomics**, 23:39. 2022
4. Thi Hai Yen Nguyen†, Tin Nguyen, **Quang-Huy Nguyen†***, Duc-Hau Le*. *Re-identification of patient subgroups in Uveal Melanoma*. **Frontiers in Oncology**, 11:731548. 2021
3. **Quang-Huy Nguyen**, Duc-Hau Le. *Similarity calculation, enrichment analysis, and ontology visualization of biomedical ontologies using UFO*. **Current Protocols in Bioinformatics**, 1:e115. 2021
2. **Quang-Huy Nguyen**, Duc-Hau Le. *Improving existing analysis pipeline to identify and analyze cancer driver genes using multi-omics data*. **Scientific Reports**, 10:20521. 2020
1. **Quang-Huy Nguyen**, Hung Nguyen, Tin Nguyen, Duc-Hau Le. *Multi-omics analysis detects novel prognostic subgroups of breast cancer*. **Frontiers in Genetics**, 11:574661. 2020

PEER-REVIEWED CONFERENCES

1. Hung Nguyen, Bang Tran, Duc Tran, **Quang-Huy Nguyen**, Duc-Hau Le, Tin Nguyen. *Disease subtyping using community detection from consensus networks*. **2020 12th International Conference on Knowledge and Systems Engineering (KSE)**, Can Tho, Vietnam, IEEE, 2020; 318–323. Oral Presentation.

PRESENTATIONS AND TALKS

1. 2024 MW CTR-IN Annual Meeting at The Westin Las Vegas Hotel & Spa, Las Vegas, NV. *Topic: A Systems-level Approach for Predicting Patient Survival using Multi-omics Data*. 2024

HONORS AND AWARDS

Certificate of merit for learning and class management - Class of 2015 **Hanoi, Vietnam**
Associated with Dai Nam University *Aug, 2020*

- Top 3% among all entrants of 2015 (nearly 110 students).
- Class president in 5 consecutive years (2015-2020).

27th Mathematics Olympic for Students **Nha Trang, Vietnam**
Issued by Vietnam Mathematical Association *Apr, 2019*

- 🏆 **Received an honorable mention** in Linear Algebra among 640 undergraduates representing 80 universities, academies, colleges across Vietnam.
- Taught myself linear algebra from scratch in preparation for the competition.
- Proudly attended the competition as the first and only representative of my university and became the first Vietnamese non-engineering student in the history of the competition.

TECHNICAL SKILLS

- **Programming:** Python, R, Matlab, JavaScript
- **Single-Cell & Omics:** Scanpy, AnnData, MuData, Seurat
- **Web Development:** HTML, CSS, RESTful APIs
- **Deep Learning & ML:** PyTorch, TensorFlow 2, scikit-learn
- **Database:** MySQL, SQLite, Oracle, MongoDB
- **Clouds & Tools:** BigQuery, AWS, Git, GitHub, Docker